PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Drivehub

Submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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CONTENTS DECLARATION

I hereby certify that the work which is being presented in the project report entitled **Drivehub** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Preeti Saini** . The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place : Rajpura

Date: 07/05/2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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We would like to express our profound gratitude and sincere appreciation to everyone who has supported and guided us throughout the development of our project, "DriveHub – A Comprehensive Car Sale/Purchase Web Application".
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ABSTRACT

Cars move through DriveHub, a place built for smooth trades between people looking to buy or sell. Security wraps every step, clarity stays front and center, navigation feels natural. Old hurdles fade when tools merge into one space - accessible, dependable, efficient. Trust grows because systems support fairness, simplicity guides each interaction. Sellers find reach, buyers discover options - all within steady digital walls.

DriveHub works by letting people put their cars up for sale while others look around and buy them using a clear, step-by-step setup. What stands out is the ability to check every car listing without needing to sign up first. Access stays open so anyone can drop in, see what's on offer, weigh different models, study specs closely - no strings attached. Jumping into ownership feels easier when details come early, freely visible right from the start.

Secure payments happen through DriveHub's built-in system, keeping money exchanges protected. Built using encryption plus identity checks, the process guards personal details while blocking scams. Once a payment finishes, a full invoice appears instantly as a PDF file. This document downloads fast, serves as official proof, yet reads clear for anyone who needs it

One way DriveHub works is by letting users explore cars openly while talking instantly with sellers. Instead of waiting, buyers see every detail through full visual tours that show exactly what the vehicle looks like. Messages flow smoothly because updates happen right away across devices. Payments stay protected using systems built for safety and clarity. Each sale creates invoices automatically so nothing gets missed later. Trust grows when people know documents are clear and unchanged. Transactions become smoother since steps link together without delays. This setup adapts easily whether few or many vehicles move through it. A fresh approach takes shape - not just another site but a working system reshaped around how people actually buy cars now.

1.1 Problem Background

Looking at how cars usually change hands, middlemen pop up everywhere. Information stays hidden, paper trails get messy. Picture this: someone wants to buy but cannot see what they are getting into. Communication gaps appear, trust slips away. Payments feel risky without clear paths forward. Now shift to the seller's side - real buyers stay out of reach, record keeping drags down effort. Jump to most websites today - they block views unless you sign up first. That wall keeps people moving elsewhere. A better way would run openly, clearly, smoothly. Ownership proof matters. So does clean paperwork. Safety sits at the core.

1.2 Objective of the Project

What DriveHub aims to do begins with building a digital space where car buying and selling feels safe, clear, and smooth. Getting straight to owners happens without detours. Transactions take place under protection layers meant to guard both sides. Paperwork like invoices shows up ready-made, no extra steps needed. Clarity around who owns what runs through every part. Access stays open but controls stay tight.

1.3 Proposed Solution / System Overview

Starting things off, DriveHub lets users browse cars freely, no sign-up needed. Picture quality stands out - every image sharp, none pixelated. Talking to sellers happens straight through built-in messaging, no detours. When a deal moves forward, payments go through protected channels, encryption active throughout. Each purchase wraps up with a PDF receipt generated on its own, ready to save or share. Car listings stay detailed, updated in real time. Smoothness comes from how tasks link together behind the scenes. Trust builds because steps are clear, nothing hidden.

1.4 Key Features or Functionality

Starting with a clear view, DriveHub shows every car photo in sharp detail. Moving through the site feels smooth since signing up is not required to look around. Talking directly to sellers happens in a private chat built right into the platform. Buying a car here works step by step, guided by straightforward tools.

Payments stay protected using encryption that keeps data sealed. Every purchase produces a PDF receipt ready to save or print later. Each part fits together, making deals visible, simple, and safe.

1.5 Tools and Technologies

The following technologies were used to develop the Drivehub platform:
React.js alongside TypeScript powers the frontend, delivering an interface that moves quickly when users interact. Interfaces shaped this way respond smoothly even under heavy use. The setup allows changes without breaking existing features
Server tasks run on Node.js, while Express.js manages how features behave plus shapes the API structure. Running behind everything, these tools sort requests using clear rules built into their design
Storing things like user details, car ads, messages, and payment logs happens safely through MongoDB. What backs up this system is a flexible database handling each piece without clutter. User info sits alongside listing data, held firmly by the same tech. Messages between buyers and sellers? They're tucked into place using that same storage method. Commissions and deals leave a trace - all kept neat inside this setup
Signing in happens through Clerk, tied to Google OAuth, keeping access smooth yet protected. Session handling runs quietly behind the scenes, avoiding hiccups during use
A fresh look comes alive through Tailwind CSS, shaping a sleek interface that works smoothly on mobile devices. Layout shifts gracefully when screen size changes, keeping clarity front and center. Visual harmony stands out without drawing attention to itself. Clean lines guide the eye where needed, never distracting. Design stays light on its feet, adapting without losing balance.
Out here, coding happens inside Visual Studio Code - this editor handles most of the work. The setup leans on it heavily, almost like a default choice without much debate. Picture every task flowing through its interface, one after another. It just sticks around because switching feels messier than staying put.
Flow of the System
The core workflow of the Drivehub system is as follows:
Signing up happens through an email address or by choosing a Google login. Accessing the system begins with confirming identity online. A person enters details to start using the service safely. Getting inside relies on verified credentials sent via mail or linked accounts. Entry unfolds after information checks out correctly. Logging in takes place once identification passes security steps.
On the homepage, someone looks around while typing what they want - results update fast as choices narrow down through sliders or keywords. Picking a brand shifts everything else on screen instantly; adjusting the cost range does too.

Starting with ownership, someone listing a vehicle enters info and papers online. That person puts up photos, specs, plus proof of registration. Once entered, the car appears on the platform for others to view. Details like mileage, condition, and pricing get shared straightaway. Paperwork such as service records or title docs goes in too. From there, interested parties might reach out directly. The original owner stays in control of responses. Availability times show clearly alongside contact options. After connections form, arrangements happen off the system.

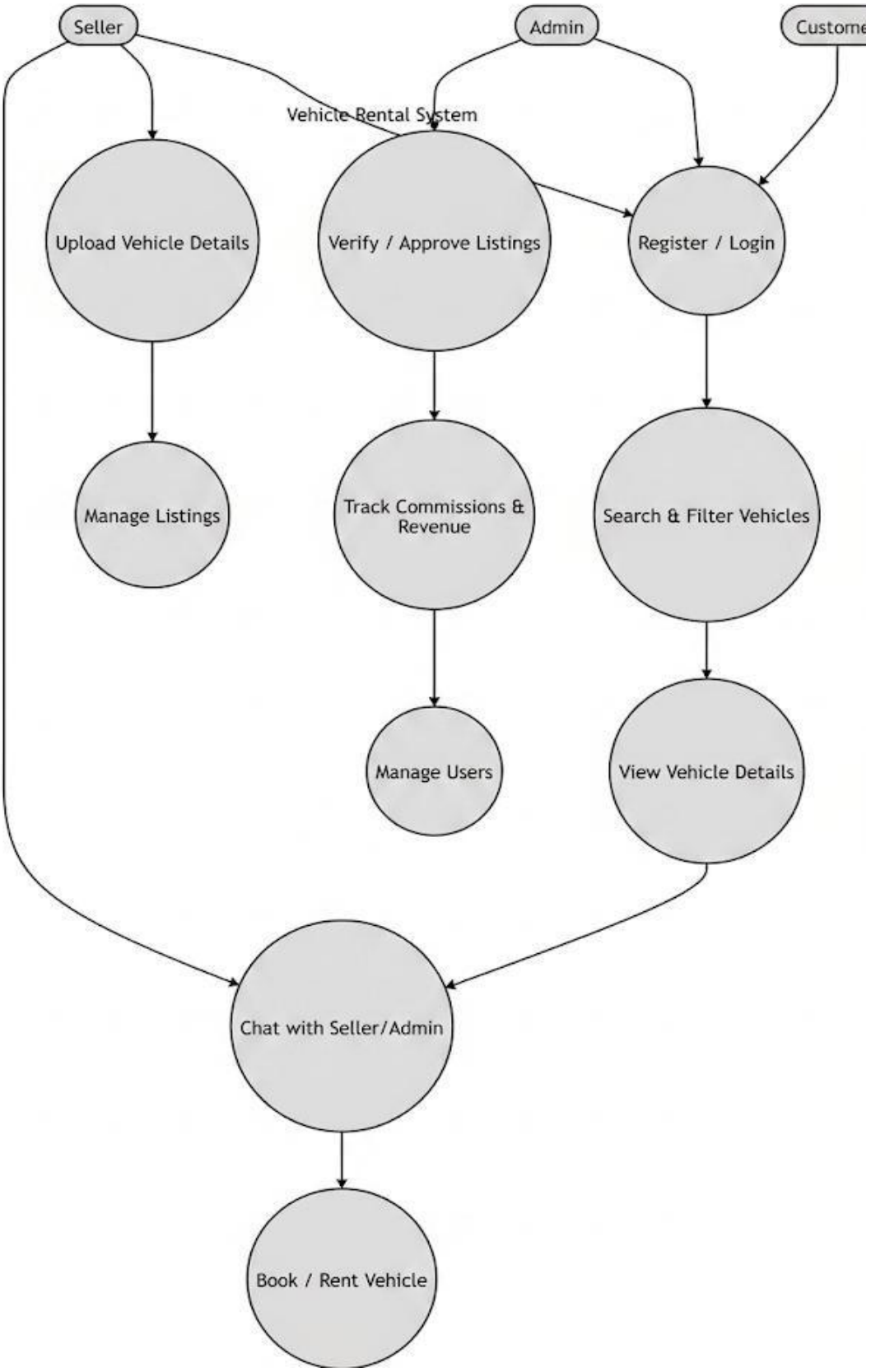
A fresh look at the upload happens when the system admin checks the vehicle details. Status confirmation comes next, done by the person managing access. Once that step finishes, commission amounts for administration are worked out automatically. Payouts for the seller follow right after, shaped by what the software determines.

Looking at a car? Get the facts straight away through built-in messaging that connects buyers right to sellers or support without delay. One moment you're browsing, next you're typing - no steps skipped. Info flows fast when questions pop up. Reach out whenever, reply comes quick. Communication stays live, always on. Details meet dialogue in one place.

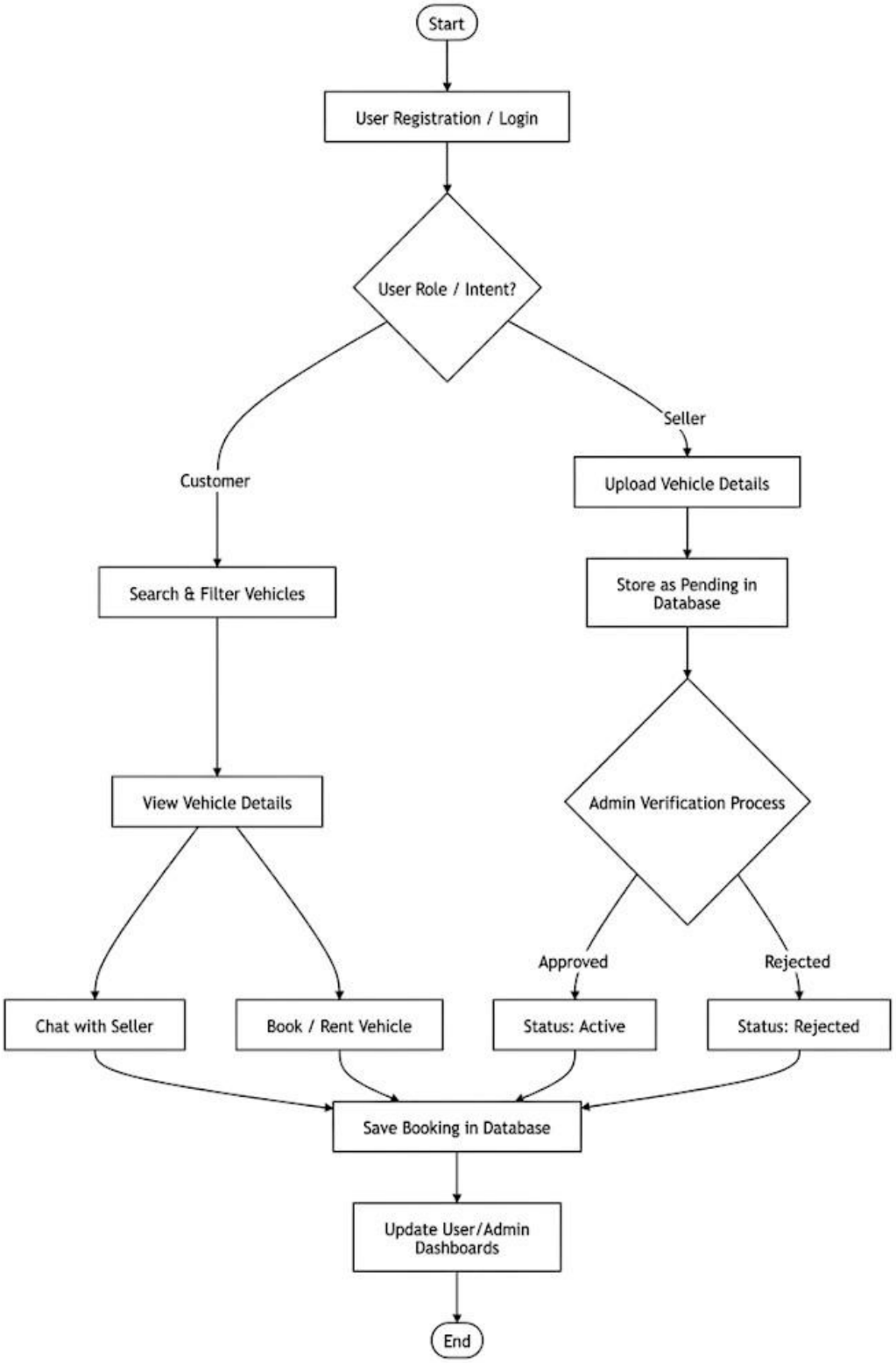
A booking happens when the buyer confirms their choice of car through the system. Once locked in, the deal moves forward without delay. Completion takes place online, where ownership steps shift into motion. The process ends cleanly after payment clears and details settle. Finalization occurs within the interface, leaving nothing pending.

From their own dashboards, users see what they've done lately, keep an eye on current listings, while income updates appear automatically. Administrators also find activity logs, live posts, plus earnings data gathered neatly in one place.

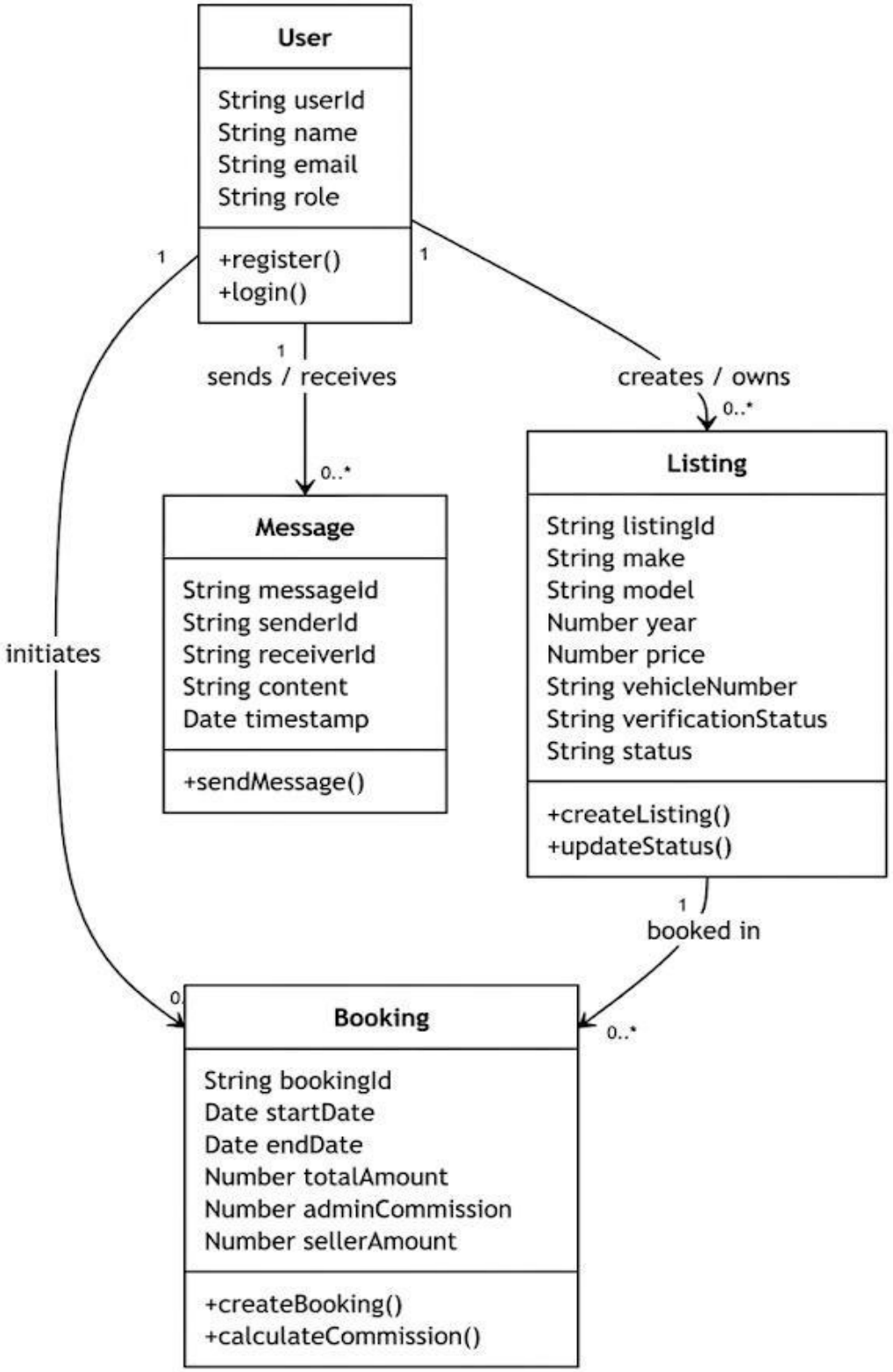
Use Case Diagram:-



Flowchart Diagram:-



UML Class Diagram:-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String email
String role
+register()
+login()
1
1
1
sends / receives
0..*
creates / owns
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Message
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String senderId
String receiverId
String content
Date timestamp
+sendMessage()
Listing
String listingId
String make
String model
Number year
Number price
String vehicleNumber
String verificationStatus
String status
+createListing()
+updateStatus()
initiates
1
booked in
0..*
Booking
String bookingId
Date startDate
Date endDate
Number totalAmount
Number adminCommission
Number sellerAmount
+createBooking()
+calculateCommission()

4. Implementation and Results

- A fresh digital workspace took shape through up-to-date coding tools across front and back ends. Built with care, the system runs on current web methods, stitching together each piece smoothly behind the scenes.
- 4.1 Implementation
- The system includes several core modules:
- User Authentication with Clerk and Google OAuth
- Managing car details becomes simpler when admins add or update info about vehicles available to rent or buy. Listing changes happen smoothly without extra steps getting in the way. Each entry stays clear and organized through regular adjustments. People who sell cars keep full control over how data appears. Updating happens quickly whenever new details need sharing
- Searching happens instantly. Pick a car by brand, type, cost, or if it's ready - right on the main screen. No waiting for new pages. Filters update the results while you stay put. Changes show up the moment you choose something
- Communication happens instantly through a chat feature right inside the platform, connecting buyers with sellers or support staff. Real-time messages flow without leaving the page, making replies quick and context clear. People talk directly, no delays, no extra apps needed - just typed words that show up immediately. This setup keeps conversation going while users stay where they are. Immediate back-and-forth helps clarify details fast, especially when decisions wait on answers. The tool lives within the interface, always ready when someone needs to reach out.
- Commissions get calculated automatically when a rental or sale goes through. This system keeps tabs on each payout behind the scenes. Every transaction is logged without manual input. Logic built into the software handles updates in real time. Payments are tied directly to completed deals. Nothing moves forward unless the deal closes first. The whole process runs on preset rules. Each step follows only after confirmation. Rentals and sales feed data into the tracker. Accuracy comes from embedded decision paths.
- Starting things off, users handle rental bookings without hassle through a clear overview of what's coming up, currently on hold, or already done. From time to time,

each reservation shows its status plainly - active ones stay visible alongside those waiting or finished. Every step flows naturally into the next, making it simple to follow along without confusion. What stands out is how straightforward planning stays when dates and vehicle choices appear together. Near the end, managing schedules feels less like work thanks to consistent updates built right in.

- Picture this - a single screen showing what matters most. Total cars ready to go sit beside current rentals, updated in real time. New users who joined lately appear just above income summaries. Everything ties together without clutter, giving clarity at a glance.

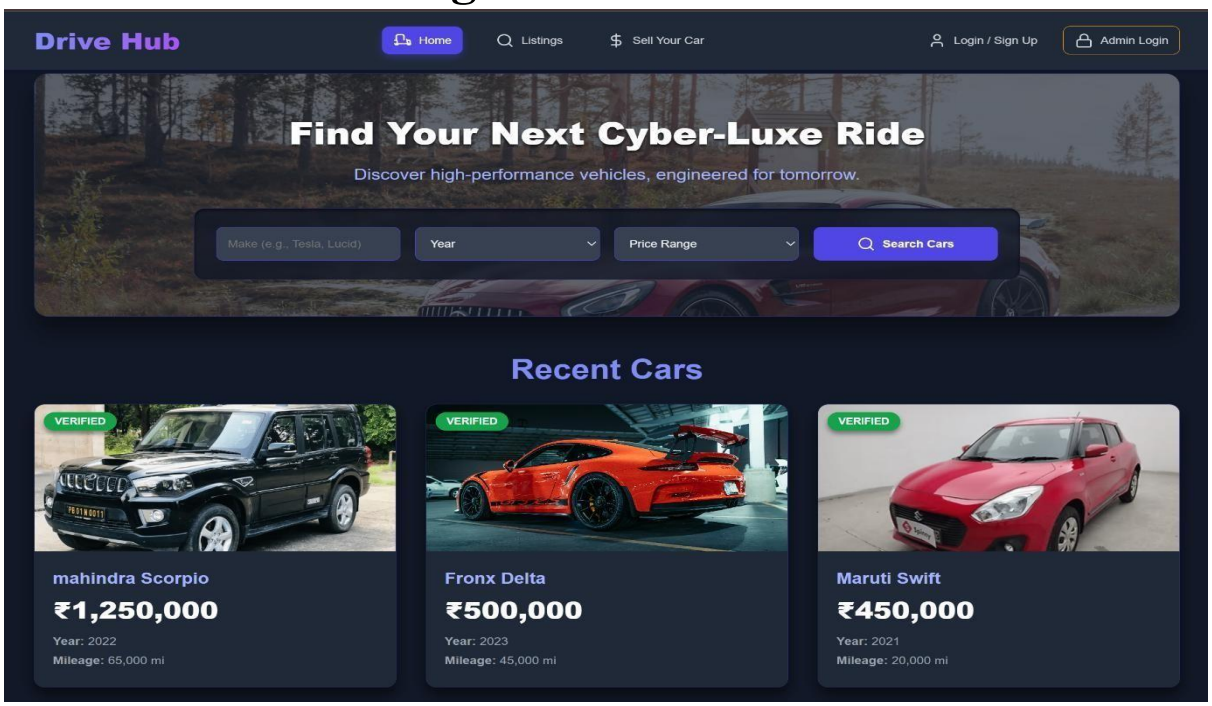
4.2 Results

A working model shows how digital tools handle car leases and purchases from start to finish. Built-in strength appears across every stage of the process. Each step connects clearly without gaps or delays. Handling transactions digitally proves reliable under real conditions.

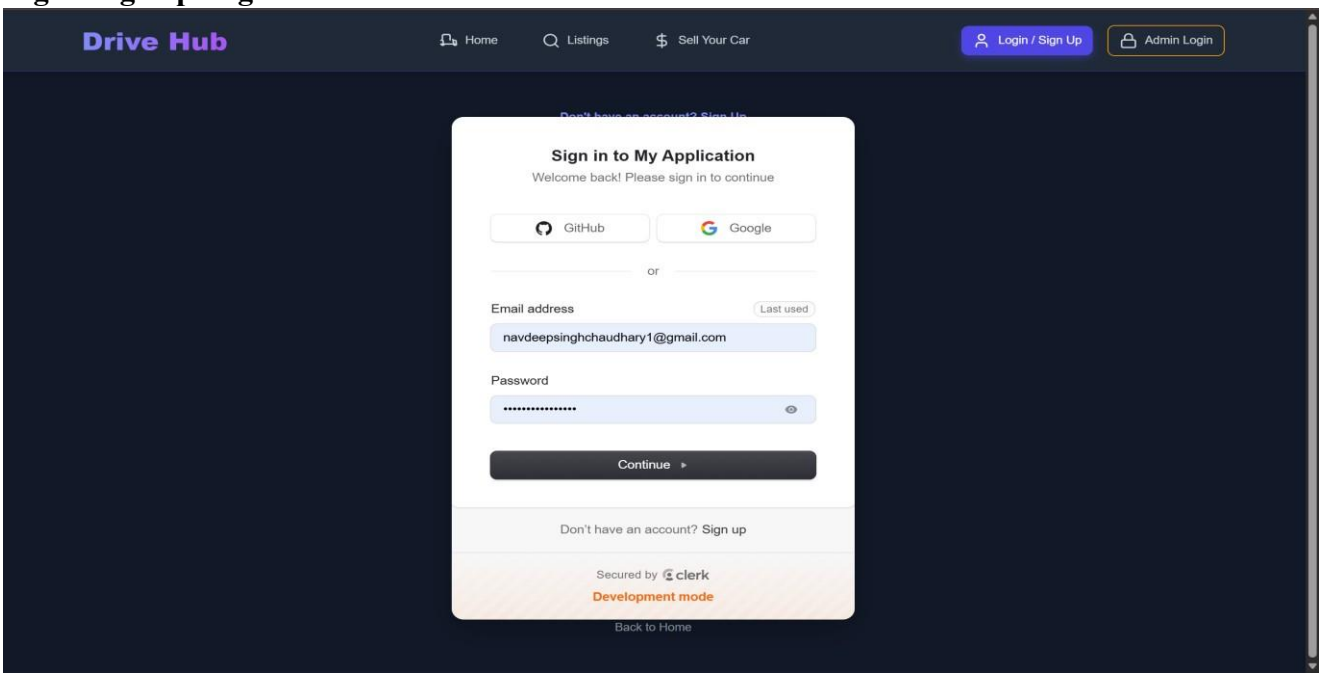
Every now and then someone finds it easier to move through the options here. Moving past basic layouts, each screen shows what matters without clutter. Clicking around feels natural because steps follow how people actually think. Information appears just when needed, never before. Behind the scenes, tools help manage everything smoothly. What you see is always up to date. Each vehicle has its own space filled with clear details. Admins keep track using a control panel that answers questions before they come up.

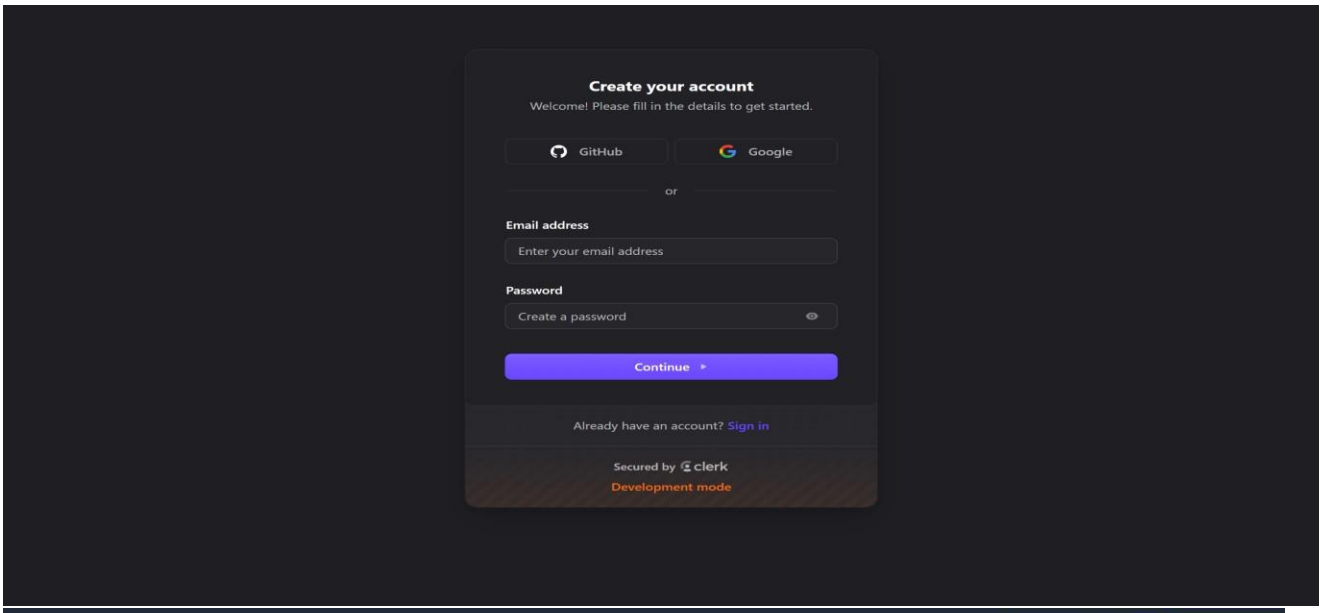
Result :-

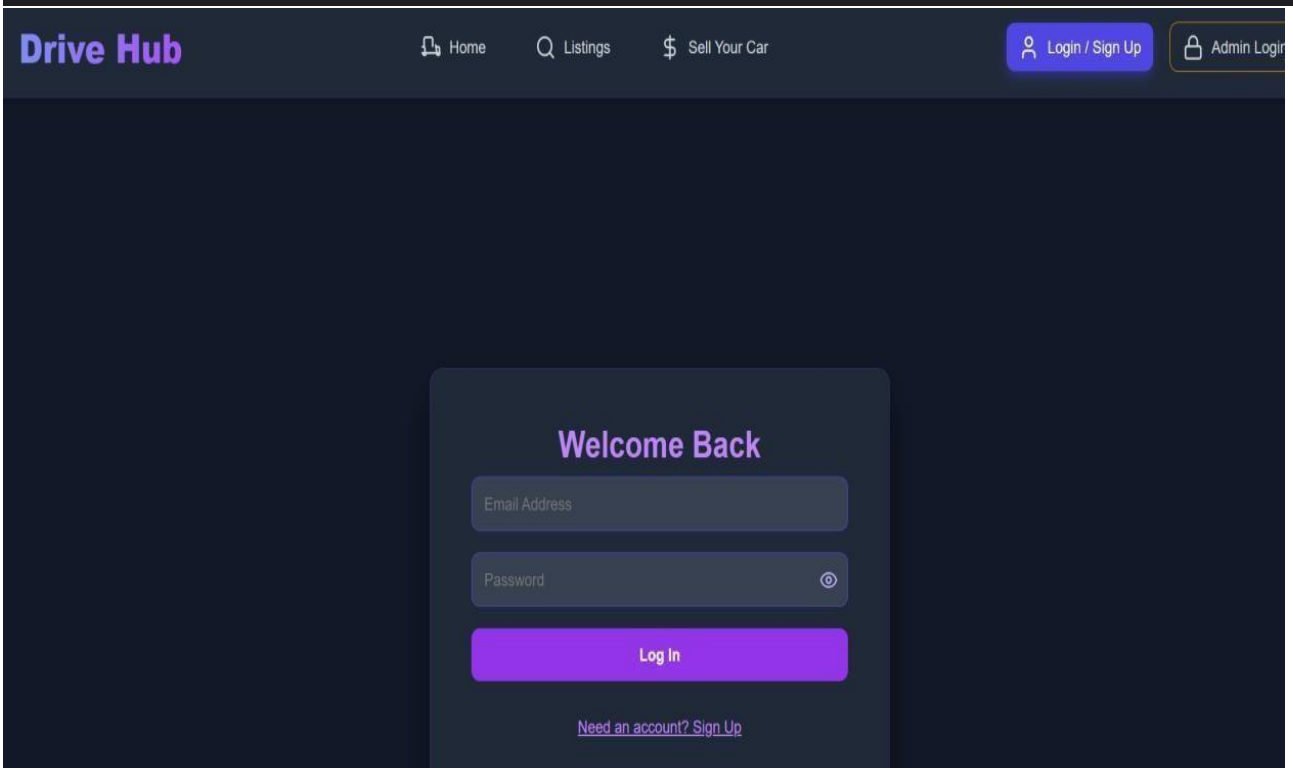
DriveHub Front Page:-



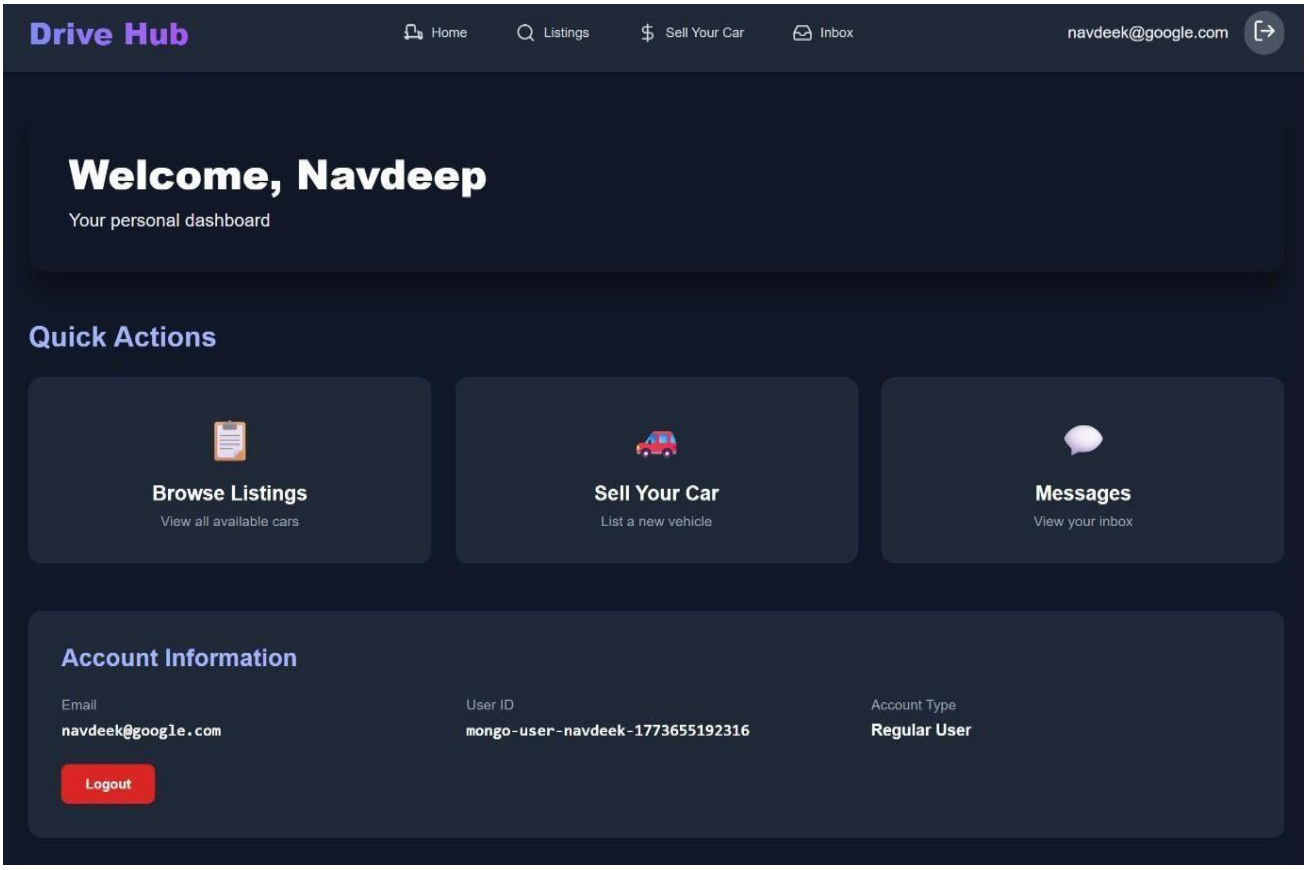
Login /SignUp Page:-



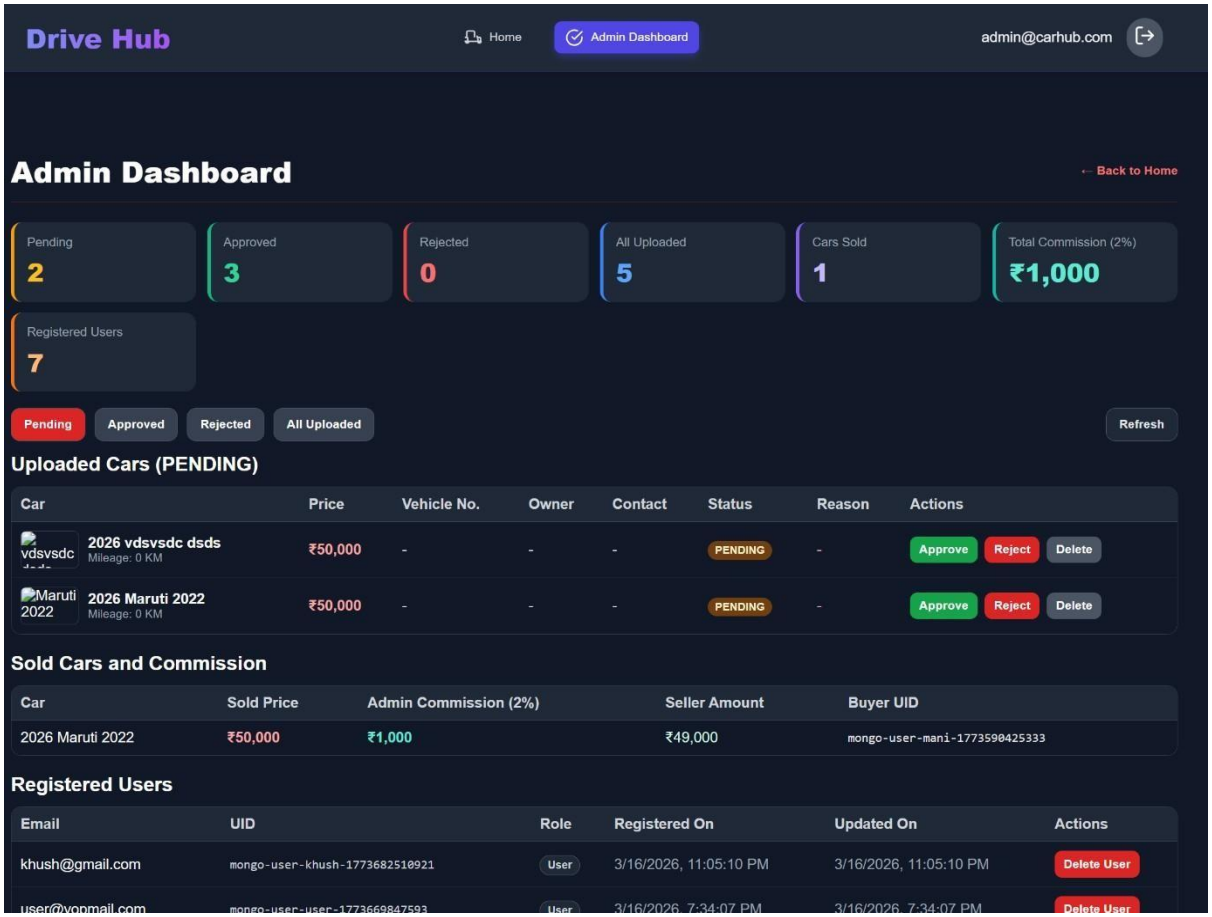

Create your account
Welcome! Please fill in the details to get started.
GitHub
Google
or
Email address
Enter your email address
Password
Create a password
Continue
Already have an account? Sign in
Secured by clerk
Development mode


Drive Hub
Home
Listings
Sell Your Car
Login / Sign Up
Admin Login
Welcome Back
Email Address
Password
Log In
Need an account? Sign Up

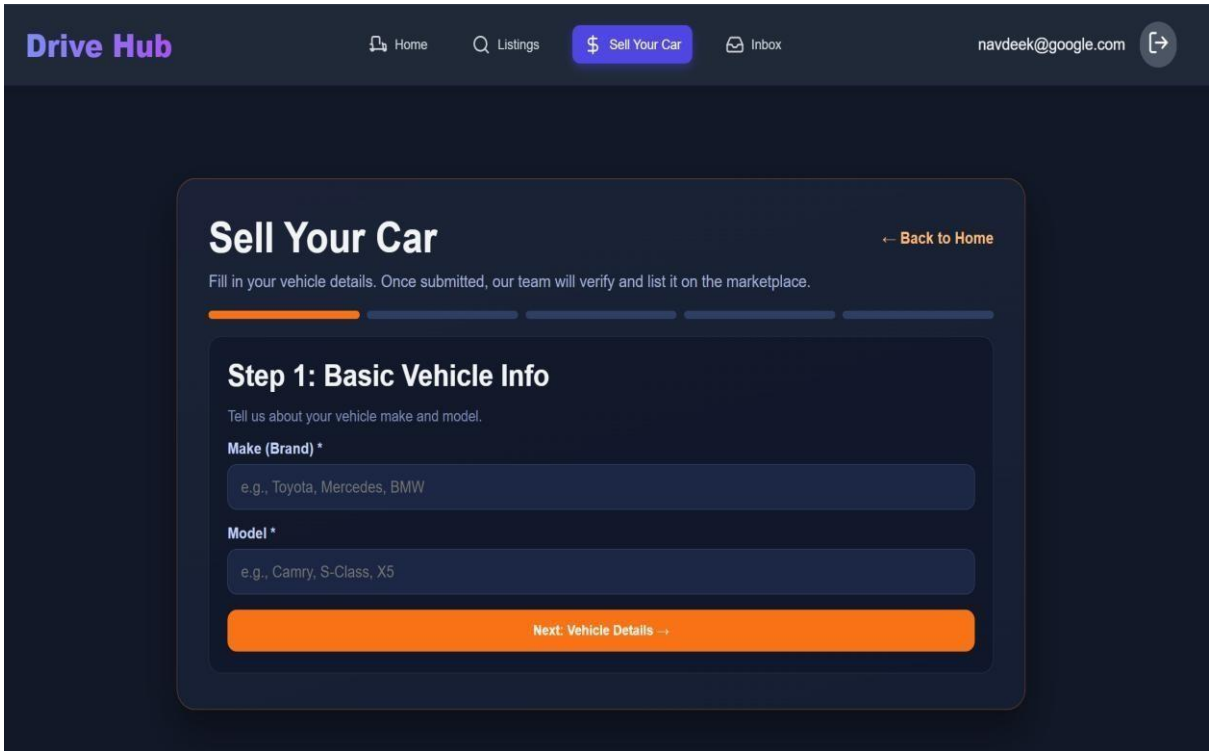
Landing Page



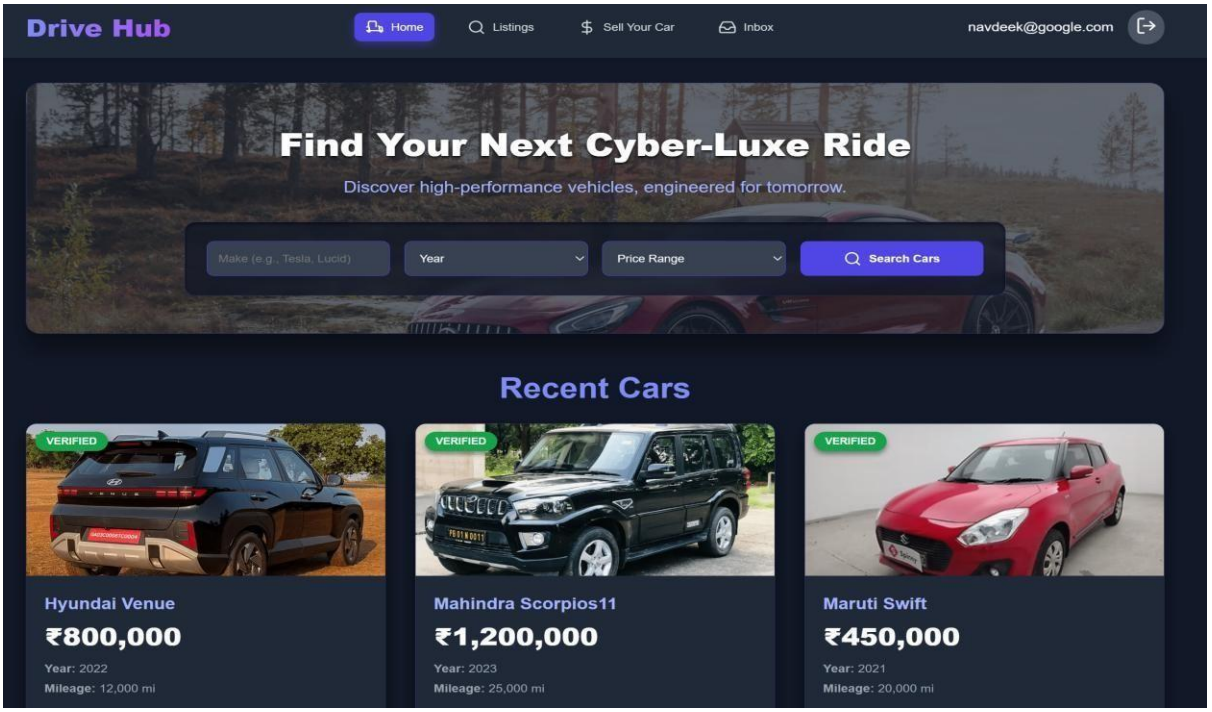
Admin Dashboard



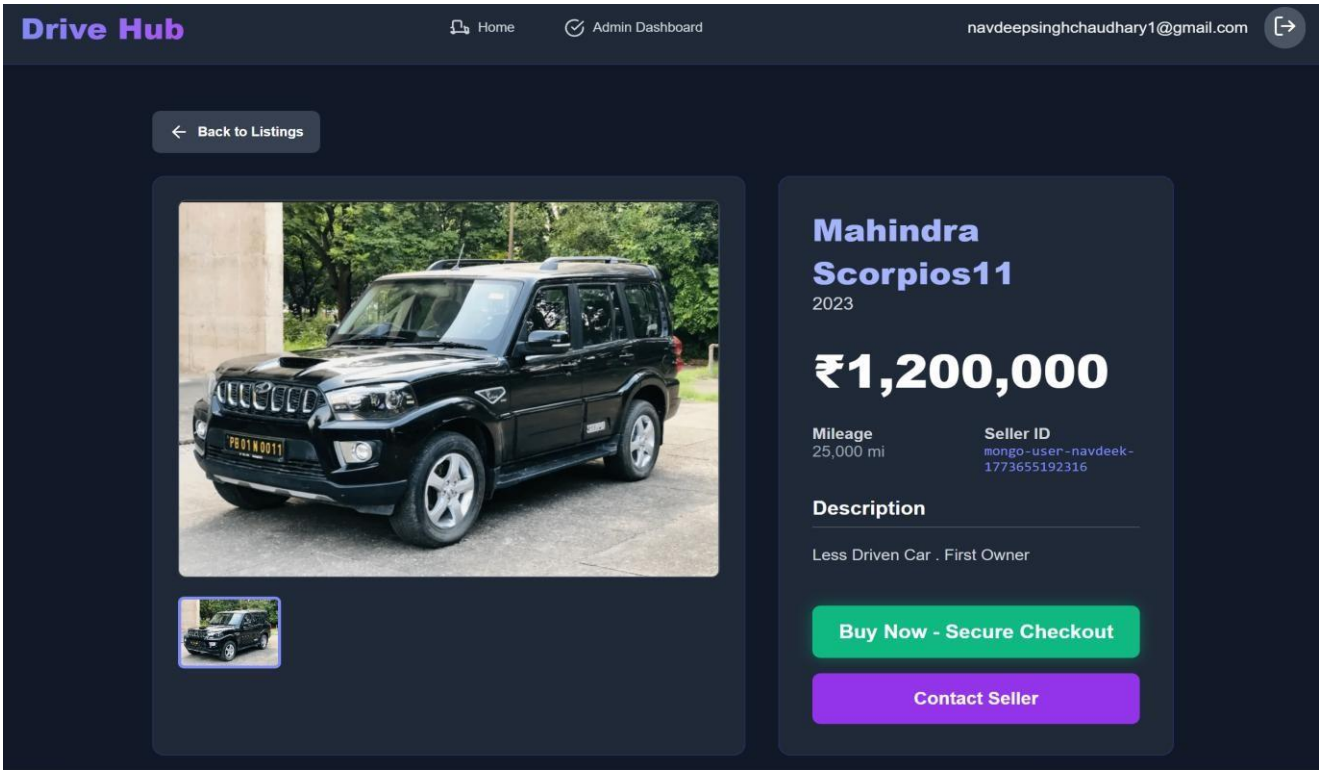
Selling Page



All Listing



Buy Car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Description
Less Driven Car . First Owner
Buy Now - Secure Checkout
Contact Seller

Final Thoughts and What Comes Next

Starts with simplicity - Drivehub builds a clear path through car rentals and sales. Not only does it link users smoothly, but also powers every step with live search tools that respond instantly. Instead of guesswork, commissions get tracked with precision behind the scenes. Security shows up quietly, making logins reliable without slowing things down. Messages flow freely between parties using built-in chats that just work. Efficiency isn't forced - it grows naturally from how pieces fit together. Admins stay in control while customers find what they need fast. Ends where others begin: ready, focused, running.

Start strong with how Drivehub simplifies finding, booking, or buying cars. Behind it, admins get solid controls for tracking stock and handling users. Built on current web frameworks, the system runs securely. Because of its stable design, scaling up feels natural. Performance stays steady even when demand climbs.

Future Work

One way forward might involve upgrading certain parts of the system later on. Some updates could come through better tools added down the line

Payments connect smoothly across borders, using modern systems that keep money transfers safe. Global purchases happen without hiccups thanks to built-in support for updated transaction tech. Security stays strong while customers pay in different countries through reliable networks.

A fresh look at how cars are doing in rentals shows up here. Numbers shift daily based on what people choose. This view digs into which models get picked most often. Movement over time becomes clear without extra clutter. Past usage links directly to current demand patterns. Trends emerge when data flows steadily through the screen.

Because you looked at certain cars before, the system picks similar ones next time. Your rental history shapes what shows up when you search again. Browsing patterns help adjust suggestions over time. Past choices influence future options behind the scenes. What you did last affects what appears now without asking.

A fresh app built just for easier reach when moving around. Smooth access comes through focused design made handy anytime. On-the-go use gets simpler because navigation feels light. Portability gains strength from tailored functions that respond fast. Effortless interaction grows out of purpose-built screens you carry. Mobility improves once tools fit your pocket naturally.

A smart chatbot goes live, handling customer questions right away while guiding users through bookings. Instant replies come online, powered by learning software that adapts over time. Help appears fast, cutting wait times without extra staff. Booking steps become smoother, guided by responses shaped from past interactions. Support runs continuously, driven by patterns the system recognizes naturally.

One step at a time, better results might show up across Drivehub's entire setup. With each update, how people interact could shift in quiet but meaningful ways. Little by little, more users may find what they need without extra effort. Changes like these tend to spread through the system, touching everything from speed to simplicity. Over time, the whole feel of using Drivehub could become smoother, almost without notice.

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